

Training Configuration and Summary:

Dataset used: Training: rssi_odom_dataset_11apr25_isolado.parquet | Experiment: rssi_odom_dataset_12apr25_isolado.parquet

model_type: LSTM
group_sec: 80
timesteps_orig: 60
batch_size: 256
min_units: 16
max_units: 256
units_step: 16
dropout_min: 0.2
dropout_max: 0.3
dropout_step: 0.1
activation_functions: ['tanh', 'relu', 'elu']
optimizers: ['adam', 'rmsprop', 'Nadam']
min_layers: 1
max_layers: 2
max_epochs: 20
hyperband_iterations: 3
group_msec: 80
samples_per_sec: 12
forecast_horizon_sec: 4
forecast_steps: 48
timesteps: 60

Dataset Parameters:

Training: $\Delta t \approx 12000.0$ ms | samples/window=1.00 | timesteps=60
Experiment: $\Delta t \approx 12000.0$ ms | samples/window=1.00 | timesteps=60

Tuning:

Trials executed: 10/10

Training Performance:

Loss: 0.0061 | MAE: 0.0605

Validation Performance:

Loss: 0.0093 | MAE: 0.0773

LSTM Multi-Step Training Report

Report Generated: 2025-09-14 20:02:49

Training Start: 2025-09-14 19:15:49

Training End: 2025-09-14 19:21:01

Training Duration: 5.19 minutes

Predictions Start: 2025-09-14 19:21:01

Predictions End: 2025-09-14 19:24:59

Predictions Duration: 3.98 minutes

Best Hyperparameters:

Number of Layers: 1

Activation: elu

Optimizer: Nadam

Dropout Rate: 0.20

MAE: 2.4261

MSE: 9.7261

RMSE: 3.1187

R2 Score: 0.7093

Multi-Step Forecast:

Forecast Steps: 48

MAE: 0.0900

MSE: 0.0120

RMSE: 0.1096

R2 Score: 0.7301

Dataset size before down-sample: 31,097 rows
Dataset size after down-sample: 4,672 rows

First 5 timestamps BEFORE:

1744464597710
1744464597720
1744464597744
1744464597757
1744464597783

...

Last 5 timestamps BEFORE:

1744464971366
1744464971377
1744464971390
1744464971402
1744464971414

First 5 timestamps AFTER:

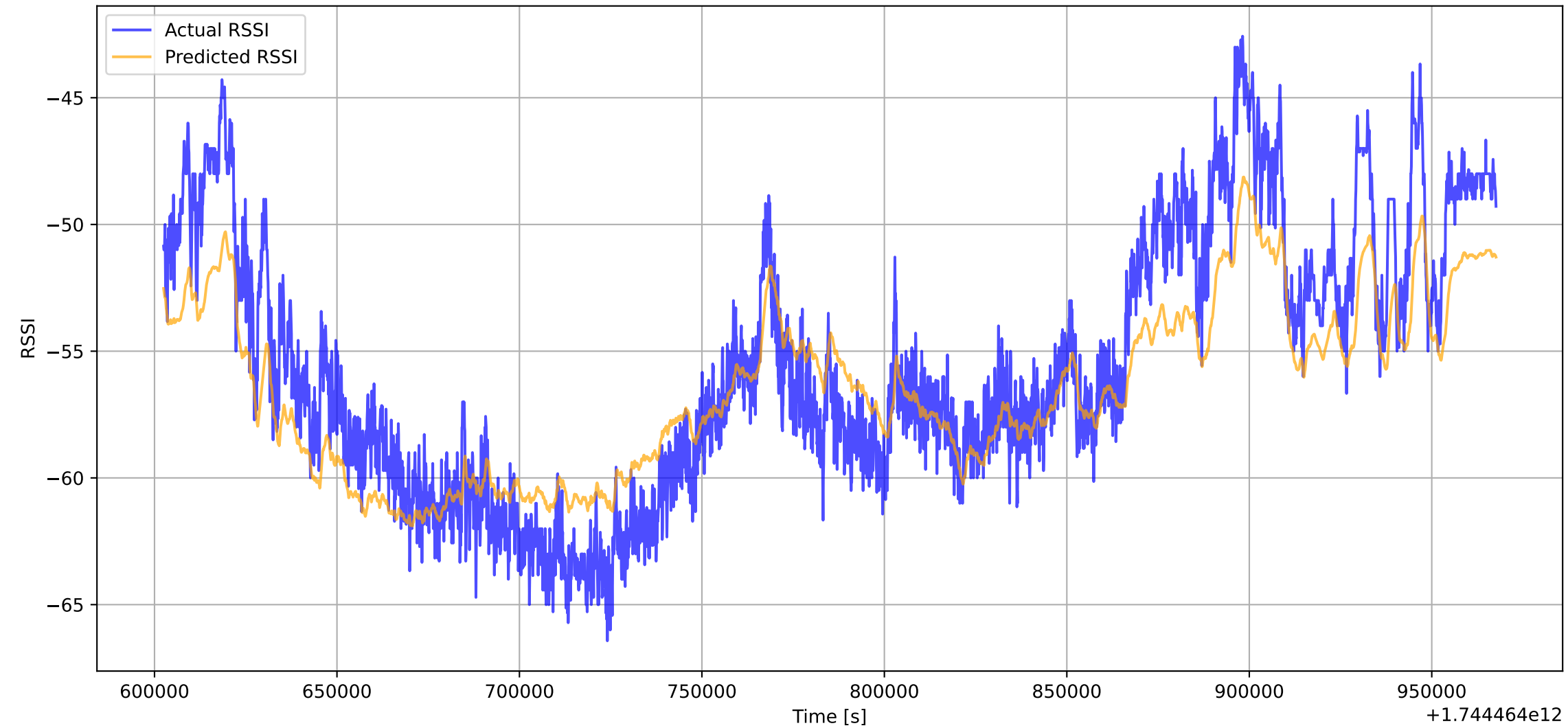
1744464597710
1744464597783
1744464597879
1744464597921
1744464598008

...

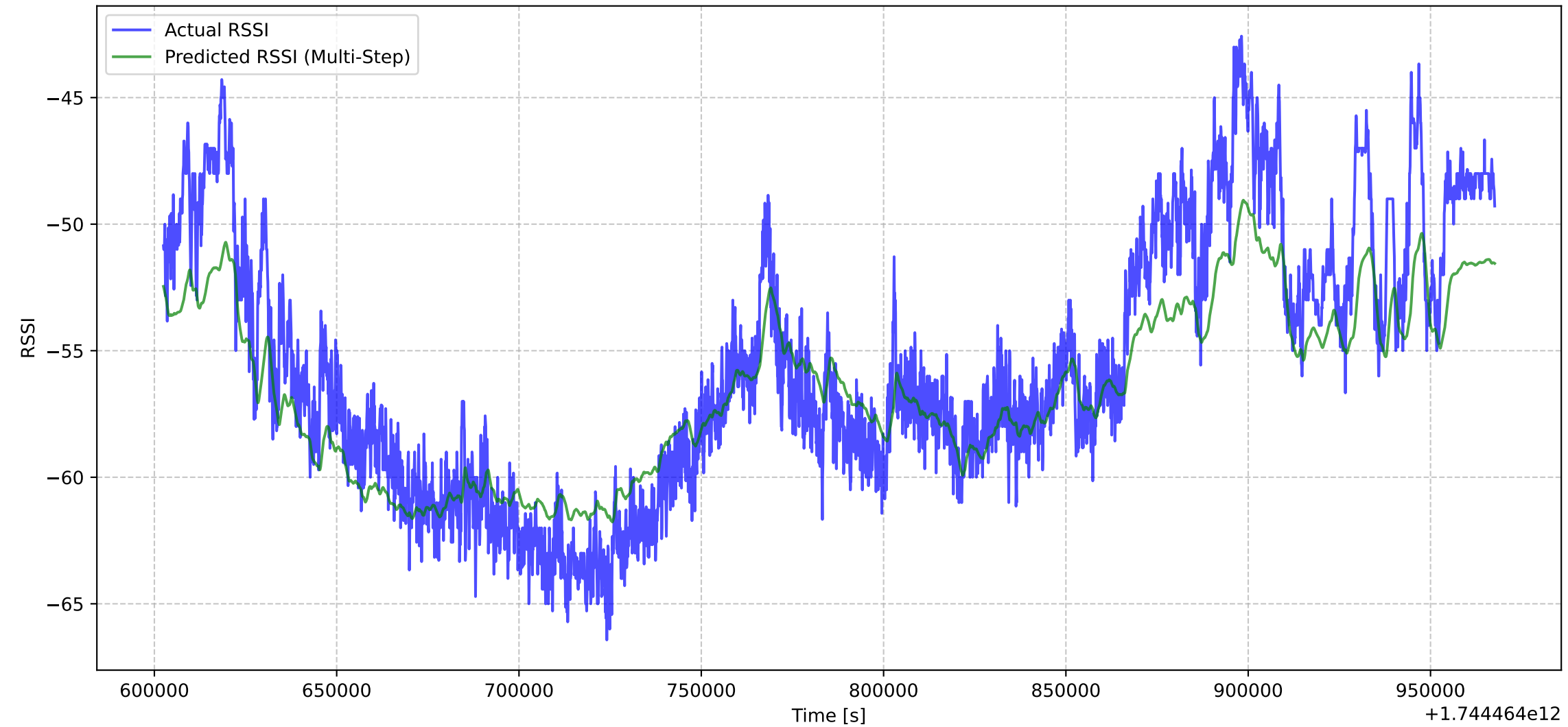
Last 5 timestamps AFTER:

1744464971041
1744464971124
1744464971209
1744464971281
1744464971366

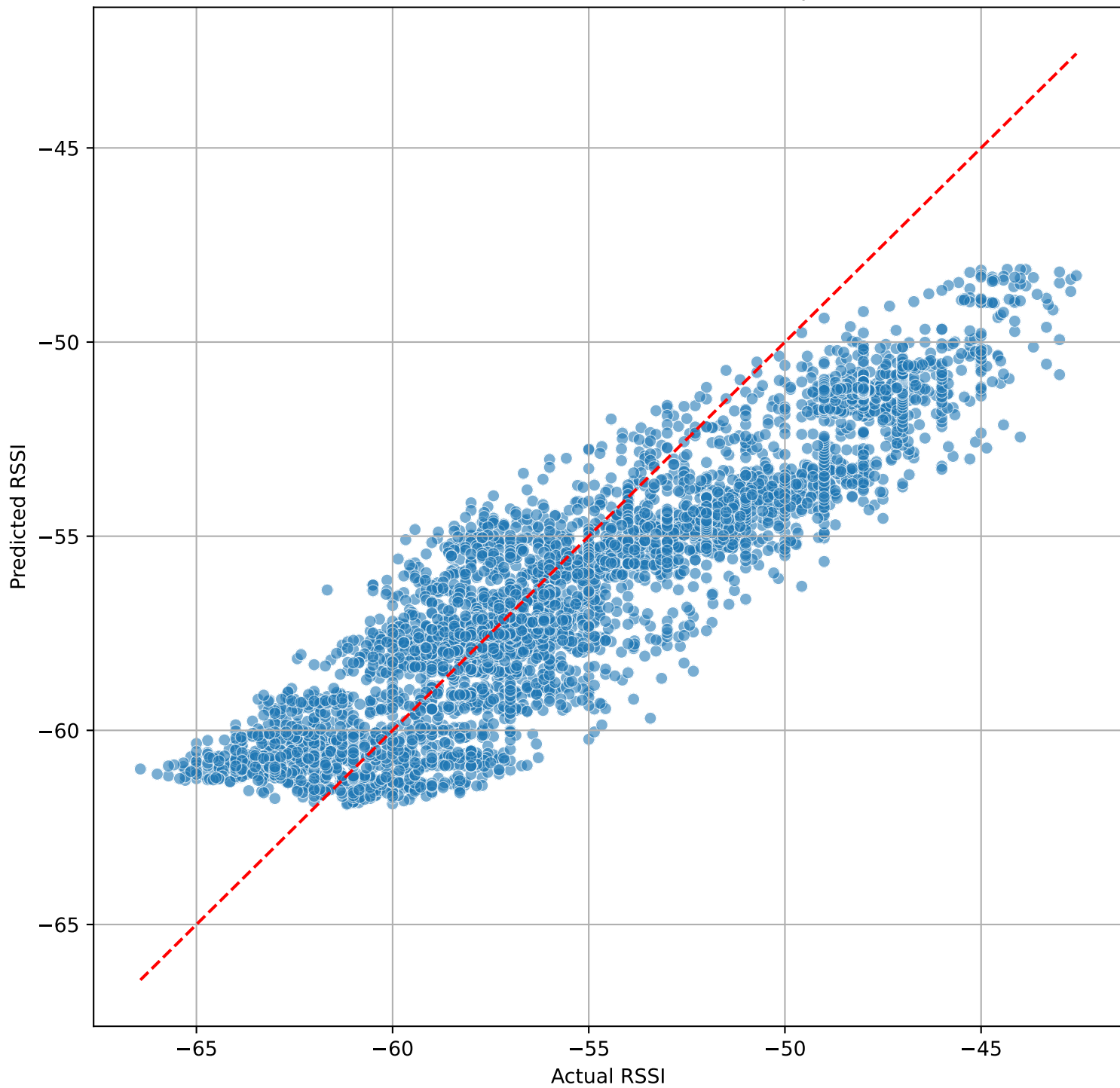
Actual vs Predicted RSSI (First Step)



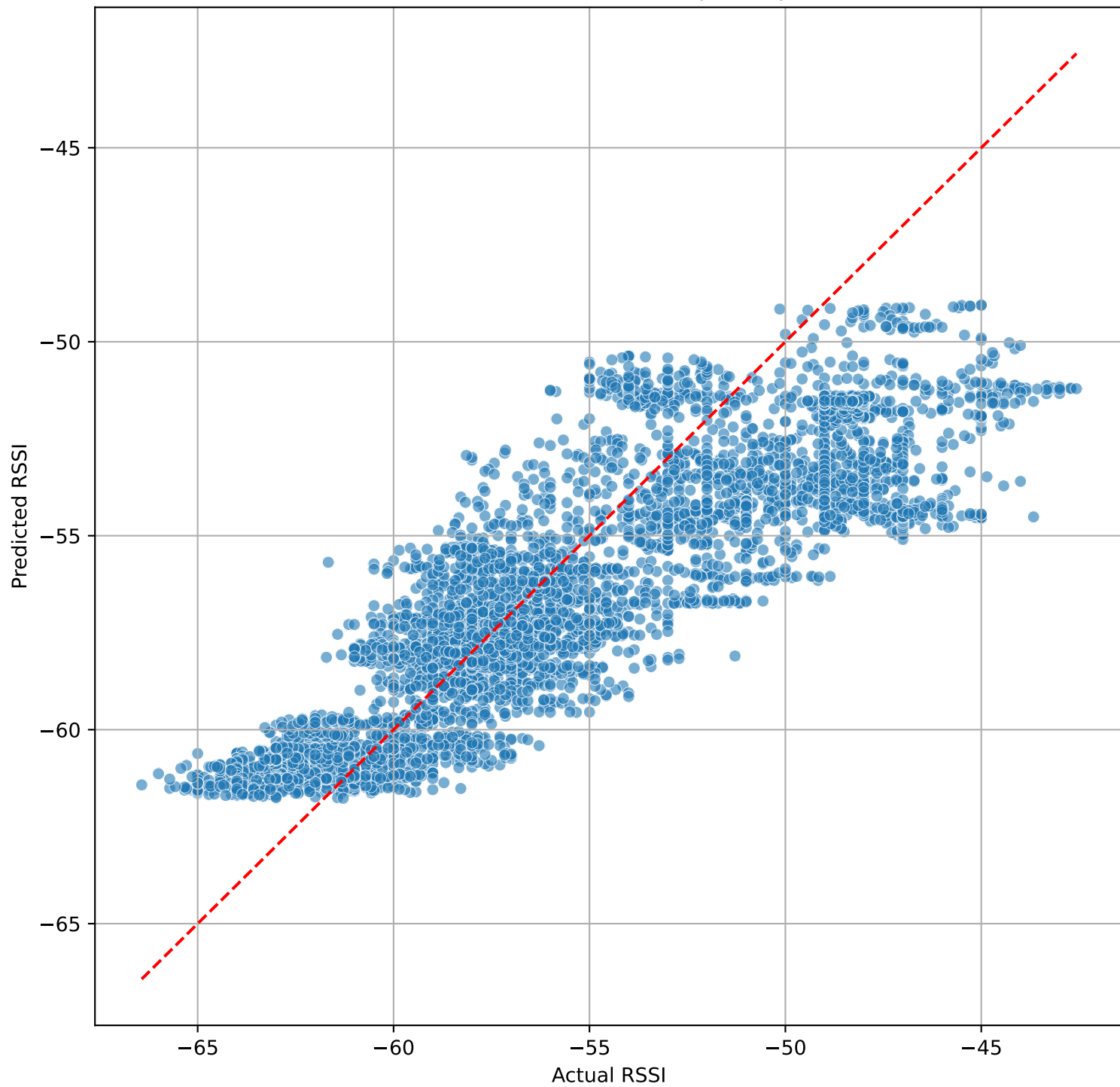
Actual vs Predicted RSSI (Multi-Step (step 48 $\approx$ 4.0 s))



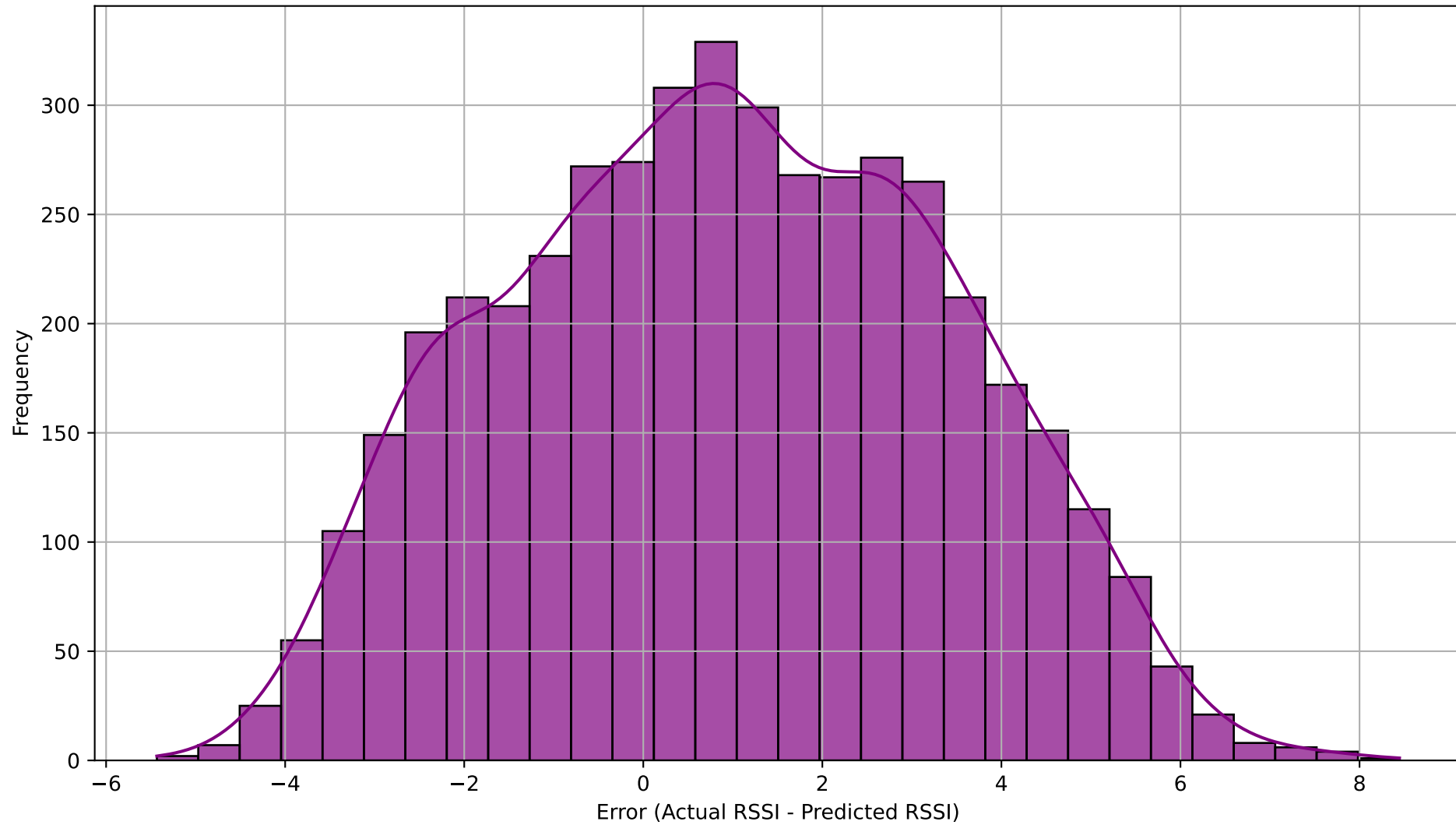
Actual vs Predicted RSSI (First Step)



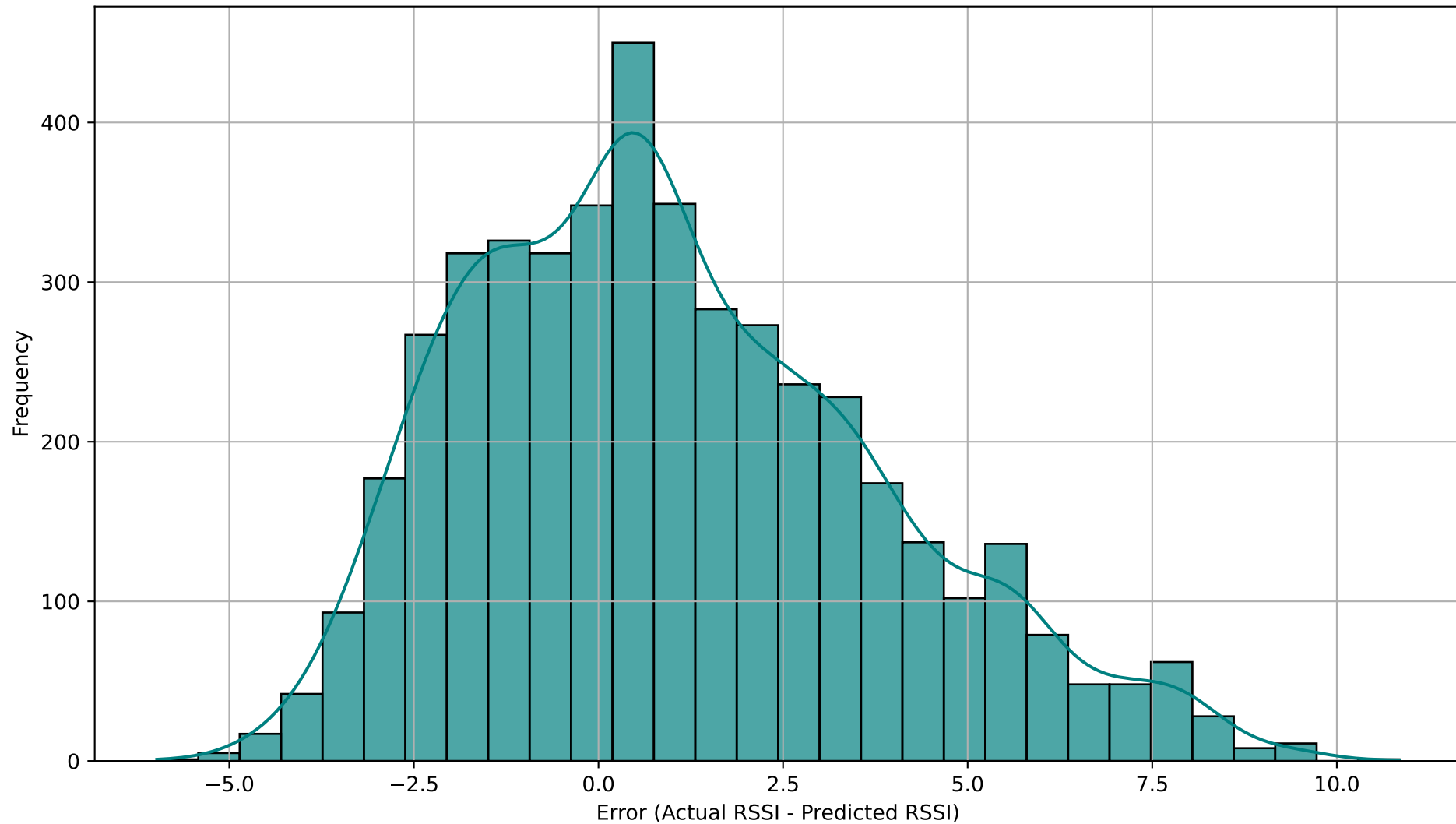
Actual vs Predicted RSSI (Last Step - step 48 $\approx$ 4.0 s)



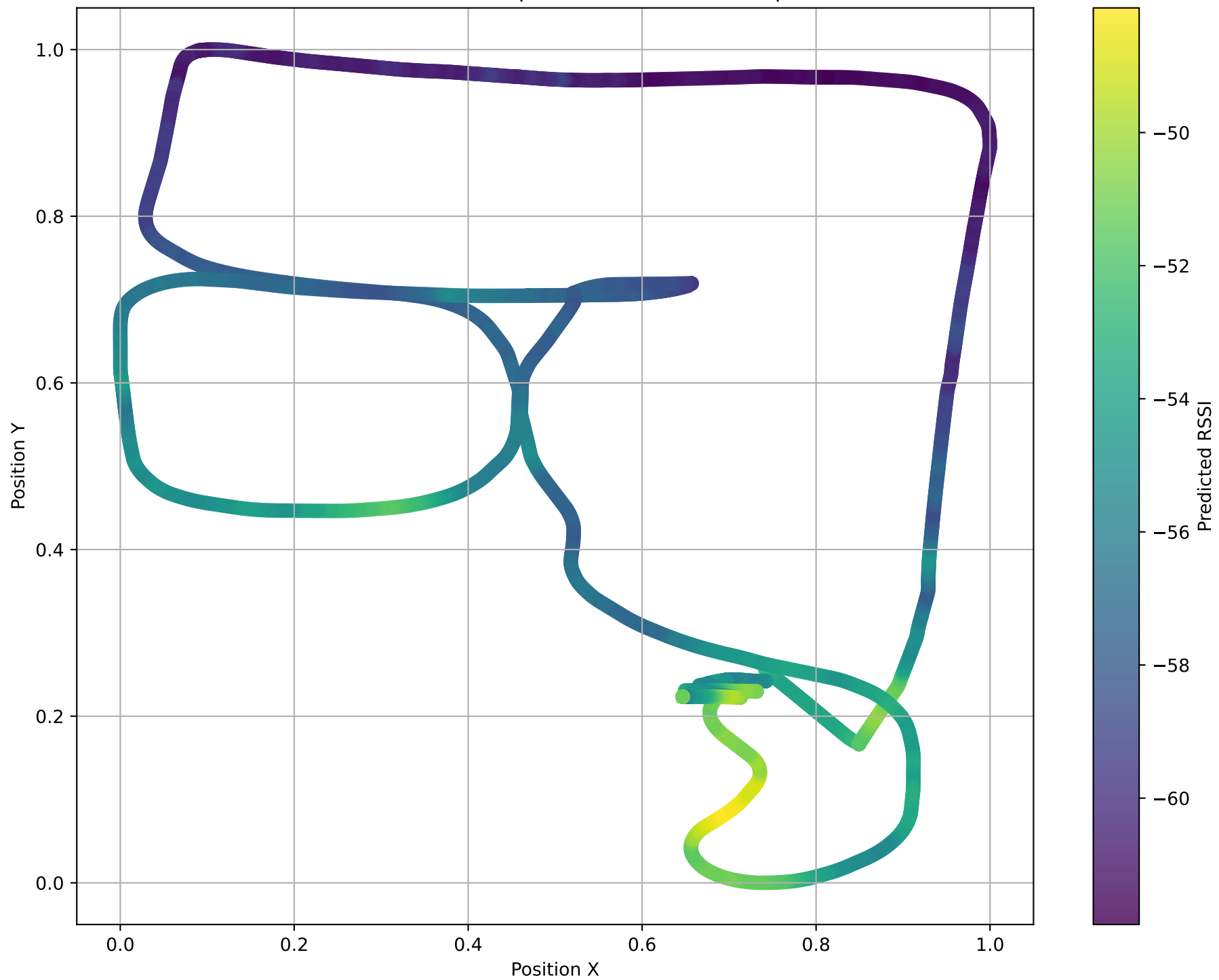
Distribution of Prediction Errors (First Step)



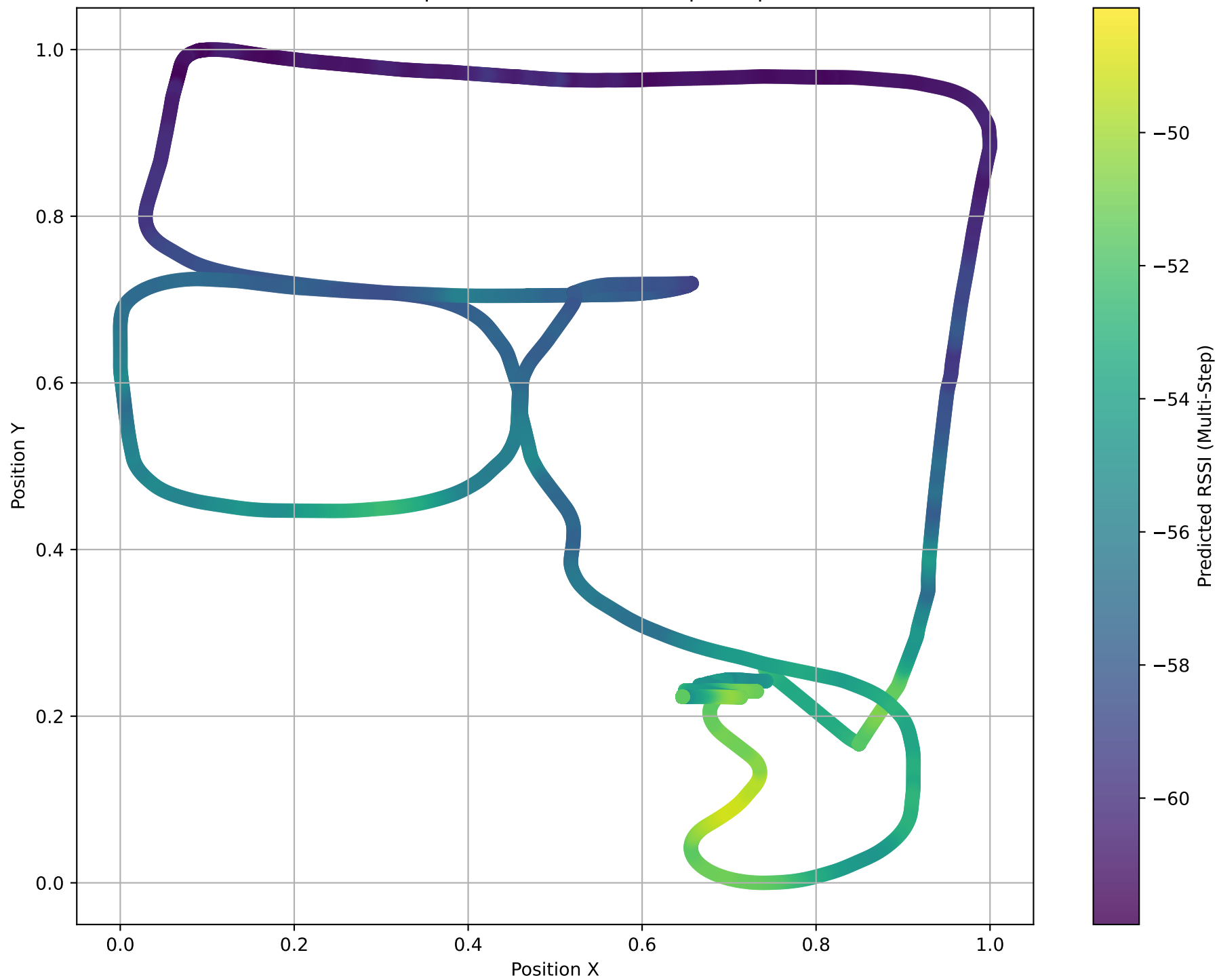
Distribution of Prediction Errors (Last Step - step 48 $\approx$ 4.0 s)



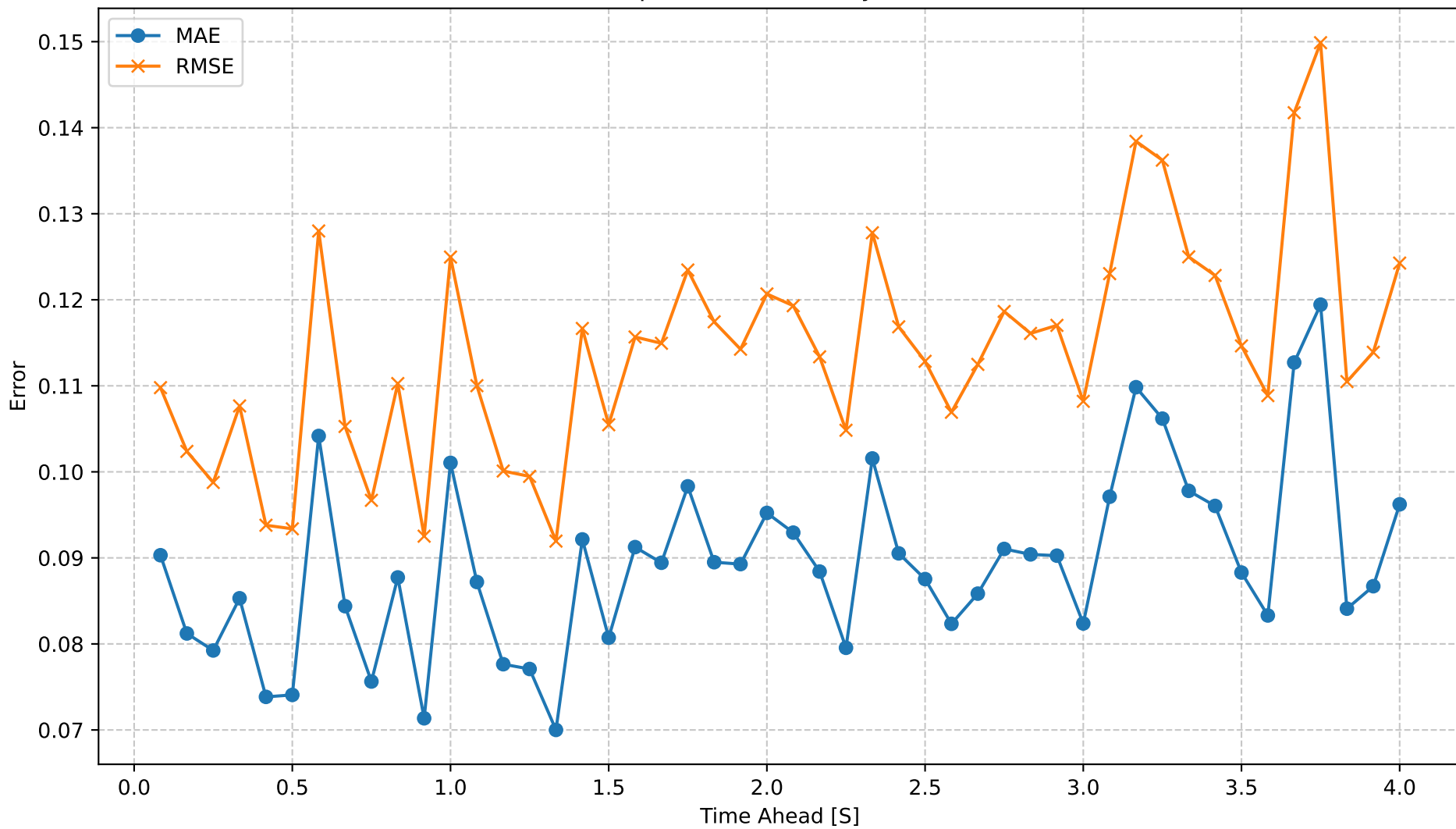
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 48 $\approx$ 4.0 s)



Multi-Step Forecast Error by Time Ahead



Resumo Automático

Dataset: Training: rssi_odom_dataset_11apr25_isolado.parquet |
Experiment: rssi_odom_dataset_12apr25_isolado.parquet
Modelo: LSTM
Forecast steps: 48

MAE / RMSE (dB)

- 1º passo : 0.09 / 0.11
- Mediana : 0.09 / 0.11
- 48º passo : 0.10 / 0.12

Crescimento do erro: 6.6% → Estável (erro cresce < 20 %)

3 piores passos (MAE): 45 (0.12), 44 (0.11), 38 (0.11)

Recomendação:

0 modelo é adequado para decisões de hand-over até o horizonte máximo configurado.